Radioactive Decay

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PHYS 2007
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1 Introduction

Write your introduction here.

2 Experimental Setup

Write your answer here.

3 Observations

4 Data Analysis

4.1 Background Radiation

For the Background Radiation, we took 100 samples of 5s duration. The result for the average/mean of backround radiation rate using python's numpy library, np.mean is: 2.68 +/- 1.54 Counts per 5 seconds. Using the python code the sample variance was given as 2.38 $(Countsper5seconds)^2$. The standard deviation is the square root of that. Here is a graph of the background radiation total counts in 5s versus how many times that happened inthe Y-axis.

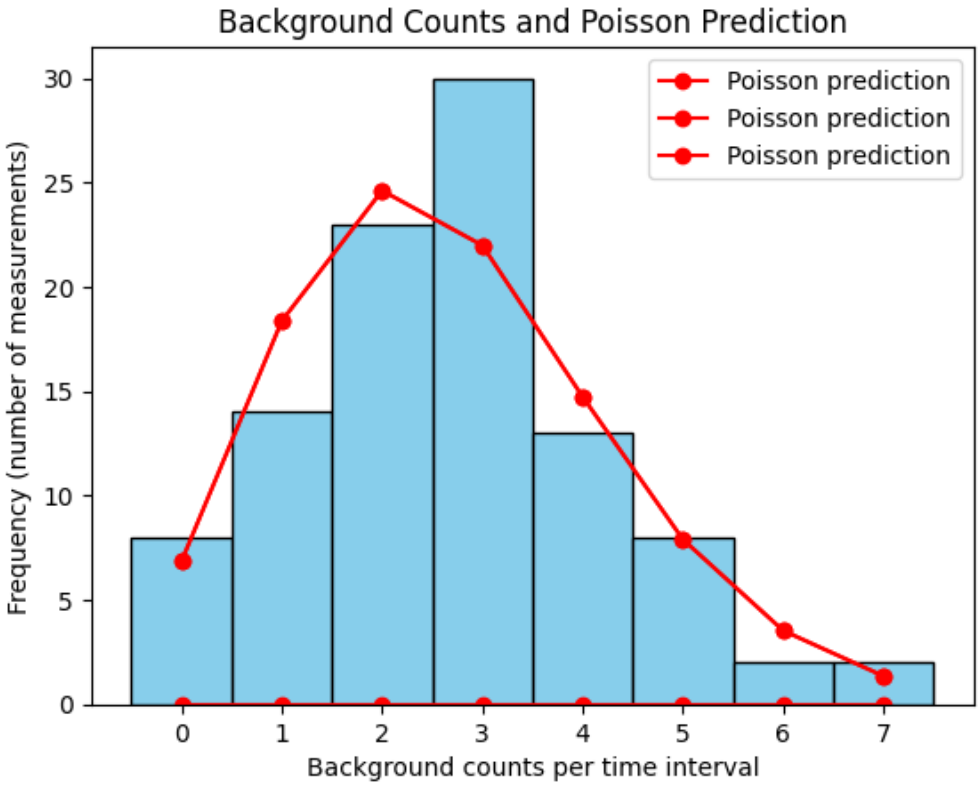


Figure 1: Radioactive decay of Barium-137m over time.

4.2 The Half-Life of Barium-137m

To find the half-life of Barium-137m, the Barium radioactive material was put very close to The GM counter. The sample rate was 5s/sample. When the data was analyzed, the background radiation counts data recorded in the previous section were subtracted easily using programming. Basically, the mean of the Background Radiation were subtracted from each sample. This is very easily done using Python. Here is the line of code used to subtract the Barium Radiation recording:

```
# 5. Subtract the corrected average background
corrected_counts = HL_corrected - bkgAvg_corrected
```

There were other calibrations to correct the data for a better fit, but the resulting change was so small that I believe that it could be neglected but still make a great fit. The Calibration is called Geiger Counter Dead Time. Here is the equation for that:

$$N_{true} = \frac{n_{observed}}{1 - n_{observed}\tau} \tag{1}$$

N_{true} is the true count, $n_{observed}$ is the observed count, and τ is the dead time of the Geiger counter.

After subtracting the background radiation, The data was plotted and here it is: The

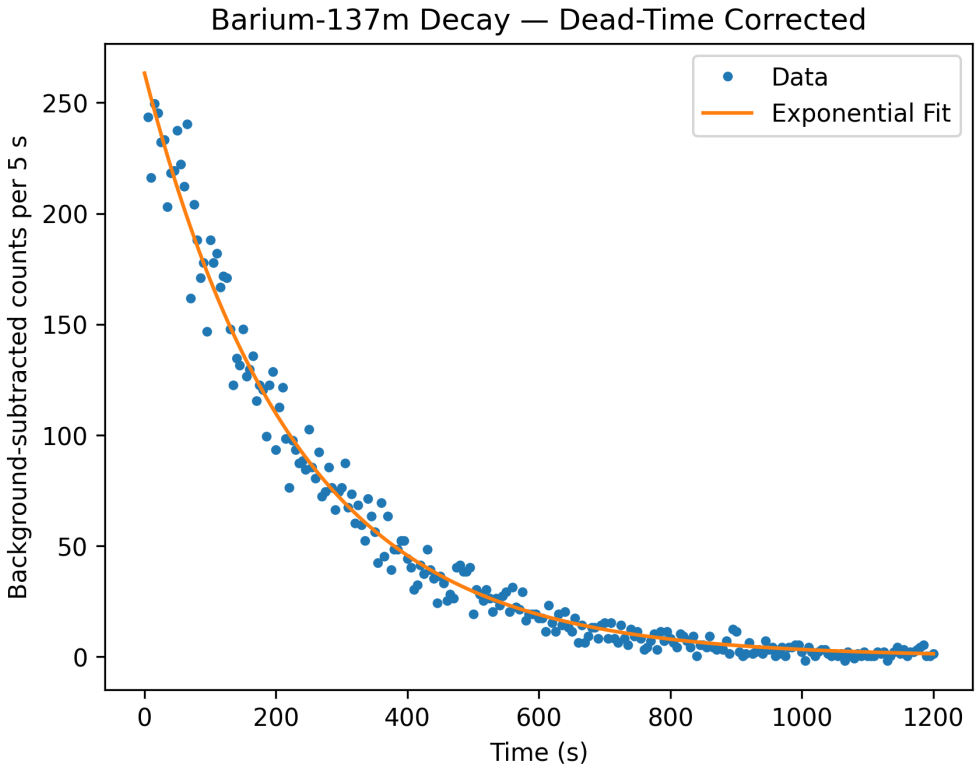


Figure 2: Radioactive decay of Barium-137m over time.

half-life of Barium-137m is: Half-Life = ln2/lambda = ln2/0.0043773832515909585 is approximately 158.35 seconds

4.3 Pb Absorption Coefficient

This part is composed of changing the thickness of the lead that the radiation has to go through and measuring what that effect has on the intensity of the radiation. The lead involved composes of different square sheets with different thicknesses but average about 1mm in thickness in zip lock bags. The data was them collected, graphed and fitted. Below is the fitted graph showing the data points and the fitted curve. The result is that the absorption coefficent is 0.02 cm^{-1}.

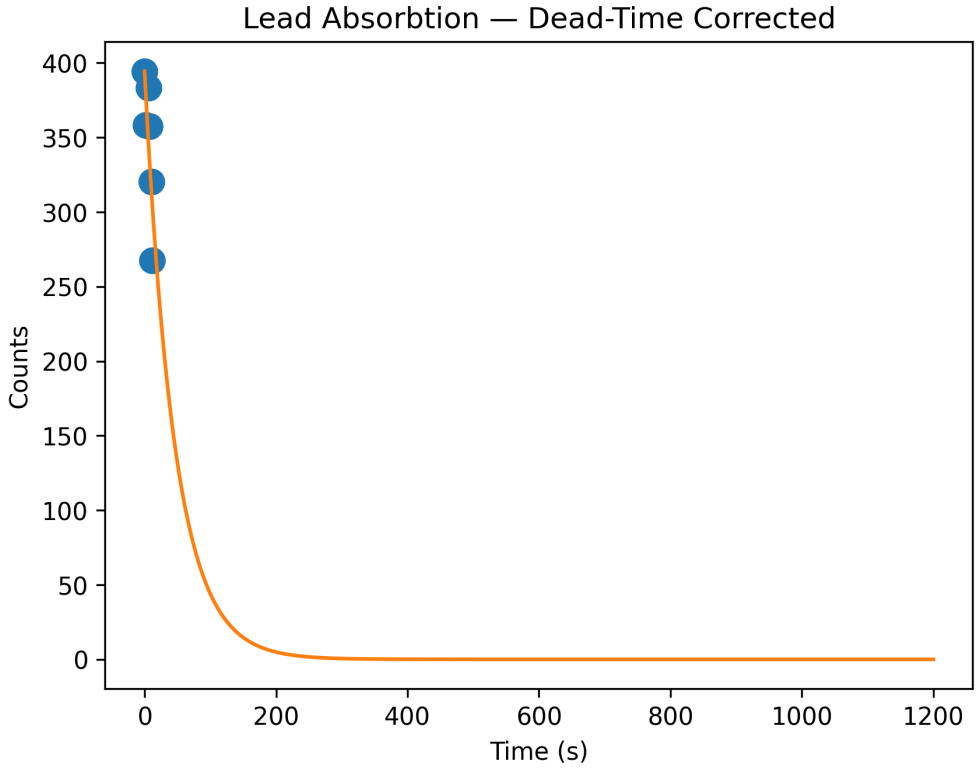


Figure 3: Pb absorption coefficient measurement.

4.4 Inverse-Square Law

The analyis for the Inverse-Square Law is very similar to previous analysis. After data is collected during the experiment, the data was sampled into a one dimensional array. Two arrays of the same size were created for graphing. The Background mean was used for correction. At that point the Curve Fit function was used. The initial function from the lab manual that I tried was:

```
def ISL(x,a,b,c,n):
    return a/(x-b)**n + c
```

After calling Curve Fit many times with no luck even after setting b, and c to zero in the call. I decided to set them to zero manually by removing them out of the equation. The new function I used was:

```
def ISL2(x,a,n):
        return a/x**n
```

Then I called:

```
popt, pcov = curve_fit(I, thickness, Counts, [1])

Where the very last parameter [1] is my
guess for u. I ended up getting N = 1.53.
```

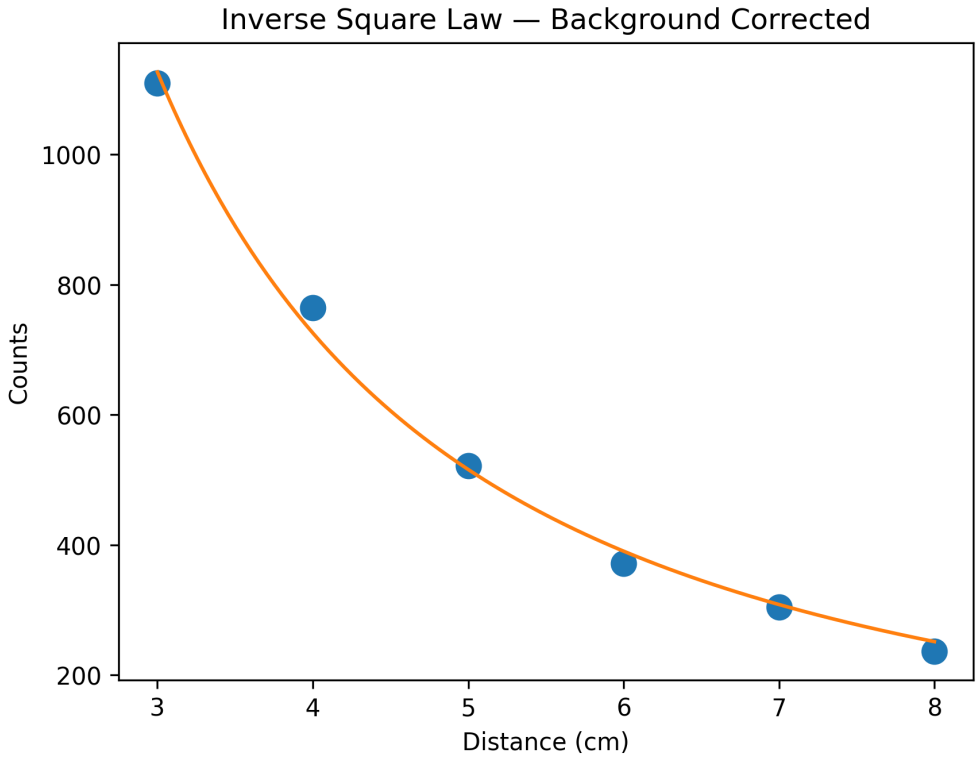


Figure 4: Inverse-square law measurement.

5 Discussion

6 Excel/Jupyter Files